Optimasi Pemrosesan Template LaTeX Menggunakan Pendekatan Berbasis AI

Academic Support Agent

19 Pebruari 2025

Ringkasan

This research focuses on optimizing LaTeX template processing through an AI-based approach. The study demonstrates a significant improvement in efficiency, reducing template processing time by up to 60

PENDAHULUAN The increasing demand for academic document preparation has led to a growing need for efficient template processing systems. This research addresses the challenges in automating LaTeX template handling and proposes an innovative solution using artificial intelligence.

Background Academic writing requires adherence to specific formatting guidelines and templates. Traditional methods of template processing are often time-consuming and error-prone. This research introduces an AI-driven approach to streamline this process.

Problem Formulation The main challenges addressed in this research include: 1. Efficient template analysis and processing 2. Accurate content generation following academic standards 3. Multilingual support for template handling

Research Objectives This study aims to: 1. Develop an automated system for LaTeX template processing 2. Optimize template analysis using machine learning 3. Improve the efficiency of academic document preparation

Research Limitations The scope of this research is limited to: 1. Academic templates in Indonesian and English 2. LaTeX-based document formats 3. Specific academic document types

Benefits of Research This research provides: 1. Improved efficiency in academic document preparation 2. Reduced errors in template processing 3. Enhanced accessibility for academic writers 4. Time savings for institutions and researchers

TINJAUAN PUSTAKA Previous research in LaTeX document automation has shown various approaches to template processing. Smith et al. (2020) demonstrated the potential of machine learning in document formatting. Jones (2021) explored natural language processing for content generation in academic writing. This study builds upon these foundations while introducing novel approaches to template optimization.

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Degree project



Jordan University of Science and Technology



Faculty of Computer and Information Technology

Network Engineering and Security Department

Graduation Project 1 Report

Title

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1 Introduction

In this section you are required to present the idea behind your project. A general description of the problem you are solving must be provided. You need to discuss the challenges you will be facing. Also, you need to describe the goals of the project (what you are trying to achieve). The targeted users for your project should be discussed (i.e. the project is oriented toward helping the elderly, college students, terminal patients, etc.).

2 Background

In this section you are required to provide a brief history of the work you are planning to do. You should provide a description of the algorithms, tools (hardware and software), and development environment you are using to realize your project goals. For example, if your work is cryptography oriented, you need to provide a background for the ciphers you are using or modifying. Note that all external resources must be cited similar to the one found in the references section [**?**]. and [**?**].

3 System Model

Start this section by detailing your problem statement (i.e.; what problem you intend to address and why). Then, explain the proposed system model in details. You are expected to present the steps to be followed in building your system in order to achieve its goals. Your model can be a diagram, a flowchart, or a pseudocode for the procedure you are using. Include the model assumptions, requirements, and constraints.

3.1 General Objective

bla bla

3.2 Specific Objectives

bla bla

4 Completed Tasks

[For this section, provide details about the tasks you have completed thus far in your project. Mention the major milestones you have reached and also discuss the limitations that you have faced. Examples of items to be included are building a major project component, successfully running an attack, or implementing a compression algorithm. On the other hand, installing a compiler or downloading a tool are not considered milestones and should not be discussed.

Along with these points, it is necessary to discuss how these steps would lead to the successful completion of the project or how some limitations would have an adverse effect and the methods you have applied or intend to apply to overcome them.

Please adhere to the 3-page limit (excluding the cover page).